

APPENDIX TO THE REQUIREMENT GATHERING DOCUMENT (RGD v2.0)

GAP-13

Appendix-M
Test Strategy & Quality Assurance
School ERP

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1. Introduction

This document is Appendix-M to the Requirement Gathering Document (RGD v2.0) for the School ERP System, addressing Gap-13: Test Strategy & Quality Assurance. It defines the overall approach, principles, levels, and processes by which the system — whose architecture, internal design, external interfaces, and deployment infrastructure were established in Appendix-I through Appendix-L — will be validated for quality before and during release.

This is a business and QA process document, not a test implementation artifact. It does not include source code, SQL, API definitions, JSON/XML payloads, test scripts, or any tool-specific automation code (Selenium, Postman, JMeter, Cypress, Playwright, or unit test code in any programming language). It describes what must be tested, at what level, by whom, and against what quality bar — not how a specific test is coded.

Per the governing rules for this task, this document does not re-analyze RGD v2.0 or regenerate any prior appendix. Every module, entity, Functional Requirement, Business Rule, Non-Functional Requirement, internal-design component, API group, and infrastructure element referenced below is reused by ID from Appendices F, H, I, J, K, and L, which remain authoritative.

1.1 Document Style Consistency

This document follows the identical formatting, section-numbering, ASCII-diagram, and traceability conventions already established in Appendix-J (LLD), Appendix-K (API Specification), and Appendix-L (Deployment Architecture) — the same document family, extended into the Test Strategy & QA domain rather than introducing a new documentation style.

1.2 Document Structure

§	Content
1-5	Introduction, purpose, scope, objectives, and guiding testing principles
6-8	Overall strategy, life cycle, and the seven test levels
9-10	Functional and non-functional testing approach
11-12	Test environment mapping and test data management
13-16	Defect management: process, severity, priority, and life cycle
17-19	Traceability, documentation, and deliverables
20-22	Entry, exit, and acceptance criteria
23-25	Risk-based testing, metrics, and quality gates

§	Content
26-27	Final validation and document closure

2. Purpose

The purpose of this Test Strategy is to establish a single, consistent quality-assurance approach across all 11 approved modules (RGD v2.0 Section 12; Appendix-E), ensuring that every Functional Requirement (Appendix-E), Business Rule (Appendix-C v1.1), and Non-Functional Requirement (Appendix-F) is verifiable before the system reaches Production (Appendix-L Section 4).

- Define the test levels (Section 8) and test types (Sections 9-10) applicable to the system.
- Define how requirements trace to test coverage (Section 17), closing the loop already opened in Appendix-B (RTM) and Appendix-D (RDM).
- Define defect management (Sections 13-16), entry/exit criteria (Sections 20-21), and quality gates (Section 25) governing progression through the environments already defined in Appendix-L Section 4 (Development, Testing, Staging, Production).
- Provide a Quality Score and Approval Status for this Test Strategy itself, consistent with the pattern already established for every prior appendix in this documentation set.

3. Scope

3.1 In Scope

- Test strategy for all 11 modules: Admission, Student Information, Attendance, Examination, Fees, Library, Transport, HR & Payroll, Communication, Reports, Administration (identical module set to Appendix-J Section 4 and Appendix-K Section 18).
- Functional testing against Appendix-E's 47 Functional Requirements and Appendix-C v1.1's 71 Business Rules.
- Non-functional testing against Appendix-F's Non-Functional Requirement catalogue (Performance, Scalability, Reliability, Availability, Security, Usability, Accessibility, Compatibility).
- Test environment strategy aligned to Appendix-L Section 4's four environments.
- Defect management, traceability, entry/exit criteria, and quality gate processes.

3.2 Out of Scope

- Test automation framework selection, test scripts, or tool-specific configuration (Selenium, Postman, JMeter, Cypress, Playwright) — remain Client/Product Decision Required and are Physical/Implementation-phase artifacts.
- Source code, SQL, API request/response payloads (JSON/XML) — consistent with the same exclusions already applied in Appendix-J, K, and L.
- Low Level Design, database schema, deployment scripts — remain the domain of Appendix-J, Appendix-H v1.1, and Appendix-L respectively, unmodified here.

4. Testing Objectives

- Verify that every one of the 47 Functional Requirements (Appendix-E) behaves per its documented Main Flow, Alternate Flow, and Exception Flow (Appendix-E Fields 10-12).
- Verify that every one of the 71 Business Rules (Appendix-C v1.1) is enforced exactly as specified, including its Validation Logic and Exception Handling (Appendix-C v1.1 Fields 14-15).
- Verify that every applicable Non-Functional Requirement target (Appendix-F) is met — response times, scalability limits, availability targets, and security controls.
- Verify that the system's internal component design (Appendix-J) and external API groups (Appendix-K) behave consistently with their documented conceptual contracts.
- Verify that the deployed system (Appendix-L) meets its operational quality bar — backup/restore, disaster recovery, monitoring/alerting — before Production release.
- Provide Project Sponsor and Management (Appendix-I Section 8 User Roles) with objective, metric-based confidence (Section 24 Test Metrics) to authorize each Quality Gate (Section 25).

4.1 Objective-to-Metric Alignment

Each objective above is measurable, not aspirational — Section 24 (Test Metrics) defines the specific measurement for Requirement Coverage, Business Rule Coverage, and NFR Compliance Rate that operationalize the first three objectives, and Section 25 (Quality Gates) defines the specific decision points where the last objective (Quality Gate authorization) is exercised. No objective in this section exists without a corresponding measurement mechanism elsewhere in this document.

5. Testing Principles

- **Risk-Based Prioritization:** testing effort is weighted toward Critical-priority Business Rules (19 of 71, per Appendix-C v1.1 Section 2.2) and High-risk Functional Requirements (Appendix-D Field 17), not distributed evenly regardless of consequence.
- **Shift-Left:** defects are sought as early as possible in the Testing Life Cycle (Section 7) — Unit and Integration Testing (Section 8) are expected to catch the majority of defects before System Testing, consistent with the layered internal design already established in Appendix-J Section 3.
- **Traceability-Driven Coverage:** no test case exists without a traceable link back to a Functional Requirement, Business Rule, or Non-Functional Requirement (Section 17) — untraceable testing is not considered part of this strategy.
- **Environment Parity:** test results carry different weight depending on environment fidelity (Appendix-L Section 4.2) — a defect not reproducible in Staging (production-representative) carries different investigation priority than one found only in Development.
- **Independent Verification:** Business-Rule-critical and Security-critical test execution (Section 23) is independently verified, not solely self-certified by the implementing developer, consistent with the Separation of Duties principle already established in Appendix-L Section 22.2.
- **Defect Prevention Over Detection:** root-cause analysis (Section 13) feeds back into Entry Criteria (Section 20) for subsequent test cycles, so recurring defect patterns are addressed structurally, not merely re-tested repeatedly.

5.1 Principle Application Summary

These six principles are not independent slogans — they compound across the Testing Life Cycle (Section 7): Shift-Left determines when testing happens, Risk-Based Prioritization determines where effort concentrates, Traceability-Driven Coverage determines what counts as done, Environment Parity determines how much weight a result carries, Independent Verification determines who signs off, and Defect Prevention determines whether the next cycle improves on the last. A test cycle that satisfies only some of these principles is not considered fully aligned with this strategy.

6. Overall Testing Strategy

The overall strategy layers test levels (Section 8) across the environment progression already established in Appendix-L Section 4.1 (Development → Testing → Staging → Production), with each level's exit criteria (Section 21) gating entry to the next.

Environment (Appendix-L Section 4)	Primary Test Levels Executed	Primary Owner
Development	Unit Testing, initial Integration Testing	Development Team (Appendix-J component owners)
Testing (QA)	Integration Testing, System Testing, Regression Testing	QA Team
Staging (UAT-equivalent)	User Acceptance Testing, Smoke Testing, Sanity Testing, non-functional validation	QA Team + Client (Project Sponsor, Appendix-I Section 8)
Production	Smoke Testing, Sanity Testing (post-release verification only)	QA Team + Infrastructure/DevOps (Appendix-L Section 22)

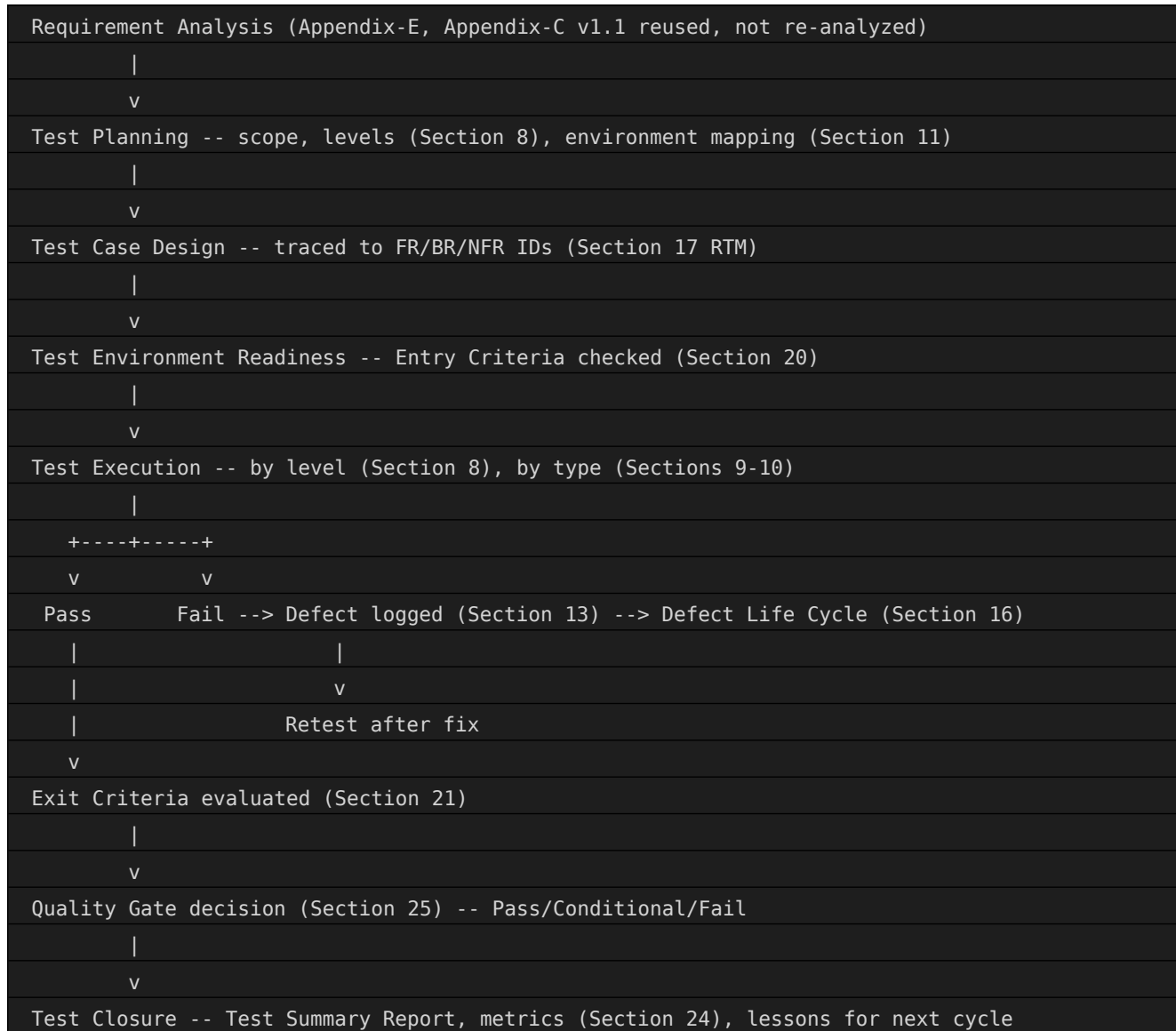
This strategy does not prescribe a specific automation-vs-manual split — that ratio, and any specific tool selection, remains Client/Product Decision Required and is an Implementation-phase decision consistent with this document's implementation-neutral scope.

6.1 Module Coverage Under This Strategy

All 11 modules (Admission, Student Information, Attendance, Examination, Fees, Library, Transport, HR & Payroll, Communication, Reports, Administration — identical to Appendix-J Section 4 and Appendix-K Section 18) are covered by every test level in Section 8, with the specific intensity weighted per Section 23's Risk Based Testing allocation. No module is exempted from any test level solely on the basis of its size or apparent simplicity — even the smallest module (e.g., Library, with only 3 entities per Appendix-G) is subject to the same Entry/Exit Criteria (Sections 20-21) as the largest.

7. Testing Life Cycle

Conceptual life cycle applied to every release cycle, consistent with the Deployment Lifecycle already established in Appendix-L Section 19.



8. Test Levels

Reuses the layered internal design (Appendix-J Sections 5-6) as the structural basis for test-level definition — each level corresponds to a boundary already established in that design.

Test Level	Purpose	Scope Boundary (per Appendix-J)	Performed By
Unit Testing	Verify a single component's logic in isolation	Individual Service/Validator component (Appendix-J Section 5)	Development Team
Integration Testing	Verify Controller-Service-Repository interaction and cross-module events	Appendix-J Section 6 interaction; Section 13 event publish/consume	Development Team / QA Team
System Testing	Verify end-to-end module behavior against Appendix-E Functional Requirements	Full module (Appendix-J Section 4.1-4.11)	QA Team
User Acceptance Testing (UAT)	Verify business scenarios against RGD v2.0 and Appendix-E acceptance criteria	Full system, business-scenario-driven	Client (Project Sponsor, Appendix-I Section 8) + QA Team
Regression Testing	Verify prior functionality remains correct after a change	Full system or affected-module subset (Appendix-D dependency-informed scope)	QA Team
Smoke Testing	Verify basic system health immediately after deployment	Critical-path flows only (Section 23)	QA Team
Sanity Testing	Verify a specific fix/change works as intended, narrowly scoped	Single defect/change area	QA Team

8.1 Test Level Sequencing

Levels are not strictly sequential in a single pass — Regression Testing recurs at every subsequent cycle once System Testing first passes, and Smoke/Sanity Testing recur at every deployment (Appendix-L Section 19), not only once per project lifecycle.

9. Functional Testing

Directly verifies the 47 Functional Requirements (Appendix-E) and 71 Business Rules (Appendix-C v1.1) already fully specified in those documents — this section does not restate their content, only the testing approach applied to them.

Test Type	Basis	Coverage Target
Requirement-Based Testing	Appendix-E Main/Alternate/Exception Flows (Fields 10-12)	100% of 47 Functional Requirements
Business Rule Testing	Appendix-C v1.1 Validation Logic (Field 14)	100% of 71 Business Rules, prioritized by Priority field (19 Critical first)
Boundary Value Testing	Appendix-G Attribute Catalogue range/format validation rules	All Mandatory=Y, range-constrained attributes
Negative/Exception Testing	Appendix-E Exception Flows; Appendix-J Section 9 Error Handling Strategy	Every documented Error Response category (Appendix-K Section 8)
Role-Based Access Testing	Appendix-E Field 38 Role Permissions; Appendix-I Section 8	Every role-to-module boundary (Appendix-I Section 8.3 matrix)
Cross-Module Workflow Testing	Appendix-D module dependency chain; Appendix-I Section 6	All confirmed cross-module relationships (Appendix-I Section 9)

9.1 Per-Module Functional Test Focus

Reuses the module-specific implementation notes already flagged in Appendix-J Section 4 as the primary functional-test focus areas.

Module	Primary Functional Test Focus
Admission	Seat/RTE-quota ceiling enforcement (BR-ADM-001, BR-ADM-003); duplicate-identity detection (BR-ADM-006)
Student Information	Mandatory-field/uniqueness validation (BR-SIS-002, BR-SIS-003); downstream-module data consistency
Attendance	Single-state-per-period enforcement (BR-ATT-001); correction-workflow routing (BR-ATT-003)
Examination	Marks range/lock enforcement (BR-EXM-002, BR-EXM-003); grade/rank calculation accuracy (BR-EXM-004)
Fees	Post-payment immutability (BR-FEE-001); duplicate-transaction prevention (BR-FEE-006); refund authorization (BR-FEE-002)
Library	Max-books/Reference-restriction enforcement (BR-LIB-001, BR-LIB-004); fine calculation accuracy

Module	Primary Functional Test Focus
Transport	Capacity enforcement (BR-TRN-001); mandatory emergency-contact validation (BR-TRN-004)
HR & Payroll	Attendance-gated payroll (BR-HR-001); access revocation on exit (BR-HR-002); duplicate-run prevention (BR-HR-003)
Communication	Relationship-scoped messaging (BR-COM-001); bulk-authorization threshold (BR-COM-002)
Reports	Read-only enforcement (BR-RPT-002); role-based field/export authorization (BR-RPT-001)
Administration	Role/Permission resolution accuracy; audit-trail completeness (Appendix-F NFR-AUDIT-001)

10. Non-Functional Testing

Directly verifies the Non-Functional Requirement categories already fully specified in Appendix-F (88 NFRs across 40 categories) — this section defines the testing approach per category, not new targets.

NFR Category	Test Approach	Source Targets
Performance	Load testing against response-time targets (page load, API, search, payment processing)	Appendix-F NFR-PERF series
Scalability	Load/stress testing at target concurrency and data volume (students, staff, transactions)	Appendix-F NFR-SCAL series
Reliability	Fault-injection testing for transaction error rate and recovery from transient failure	Appendix-F NFR-REL series
Availability	Uptime monitoring validation and planned-maintenance-window verification	Appendix-F NFR-AVAIL series
Security	Penetration testing, automated scanning (CSRF/XSS/SQL Injection), access-control verification	Appendix-F Security/Authentication/Authorization/Encryption series
Usability	Task-completion-time studies with representative Admin Staff/Teacher users	Appendix-F NFR-USA series
Accessibility	Automated (axe/WAVE-class) scan plus manual screen-reader/keyboard-navigation testing	Appendix-F NFR-ACC series (WCAG 2.1 AA)
Compatibility	Cross-browser and cross-device manual QA against the supported matrix	Appendix-F NFR-BRC-001, NFR-MOB-001

10.1 Non-Functional Testing Environment

Performance, Scalability, and Reliability testing require production-representative data volume and topology — conducted primarily in the Staging environment (Appendix-L Section 4.2), not Development or Testing, consistent with Appendix-L's own Environment Parity principle.

10.2 Selected NFR Targets Under Test

NFR ID	Target	Test Type
NFR-PERF-001	Page load ≤3 seconds (95th percentile)	Performance
NFR-PERF-007	Fee payment processing ≤5 seconds	Performance
NFR-SCAL-001	2,000 concurrent users (Client/Product Decision Required)	Scalability
NFR-AVAIL-001	99.5% uptime, academic operating hours	Availability

NFR ID	Target	Test Type
NFR-REL-001	<0.1% critical-transaction error rate	Reliability
NFR-SEC-001/002/003	Zero exploitable CSRF/XSS/SQL Injection vulnerabilities	Security
NFR-ACC-001	WCAG 2.1 Level AA conformance	Accessibility

11. Test Environment Strategy

Directly reuses the four environments already defined in Appendix-L Section 4 — this document does not define new environments, only the testing activity mapped to each.

Development (Appendix-L Sec 4)		
Unit Testing, developer-run Integration Testing		
v		
Testing / QA (Appendix-L Sec 4)		
Integration Testing, System Testing, initial Regression Testing		
v		
UAT / Staging (Appendix-L Sec 4, UAT-equivalent)		
User Acceptance Testing, Non-Functional Testing (Section 10),		
Smoke/Sanity Testing rehearsal, Disaster Recovery drill		
participation (Appendix-L Section 13.1)		
v		
Production Verification (Appendix-L Sec 4)		
Post-release Smoke Testing, Sanity Testing only --		
no exploratory or destructive testing against live data		
Environment	Test Data Sensitivity	Testing Constraint
Development	Synthetic/anonymized only (Appendix-L Section 4)	No Production-realistic scale required
Testing (QA)	Synthetic test data	Sufficient for functional coverage, not full-scale load testing
UAT/Staging	Anonymized production-like or client-provided sample data (Client/Product Decision Required, per Appendix-L Section 4)	Full Non-Functional Testing (Section 10) conducted here
Production Verification	Real data	Read-only/non-destructive verification only — no test data creation against live records

11.1 Environment-to-Test-Level Mapping Rationale

This mapping is not arbitrary — it follows directly from the Environment Parity and Environment Promotion principles already established in Appendix-L Sections 4.1-4.2. Non-Functional Testing (Section 10) specifically requires Staging rather than Testing/QA because Scalability and Performance targets (Appendix-F) can only be meaningfully validated

against production-representative topology and data volume — a Testing/QA environment sized only for functional-correctness validation would produce misleading Performance results if used for this purpose.

12. Test Data Management

Reuses the entity structure already defined in Appendix-G/H (39 entities) as the basis for test data design — this section does not invent new data structures.

- Synthetic test data is generated to satisfy every Mandatory=Y attribute and Unique Constraint already catalogued in Appendix-G's Attribute Catalogue, ensuring test records are valid per the same rules Production data must satisfy.
- Test data volume for Non-Functional Testing (Section 10) is scaled to approximate the targets in Appendix-F's Scalability NFR series (e.g., 20,000 Student records, 10 million+ AttendanceRecord entries) — exact generation approach Client/Product Decision Required.
- Sensitive/PII test data (per Appendix-G/H's Sensitive Data Matrix and PII Matrix) is either fully synthetic or anonymized from a Production-derived source — real, identifiable student/staff/financial data is never used outside the Production environment (Appendix-L Section 4 table), consistent with Appendix-F NFR-PRIV-001.
- Test data supporting Business Rule verification (Section 9) specifically includes edge-case and boundary records — e.g., a SeatAllocation record at exactly the RTE-quota ceiling (BR-ADM-003), an Invoice at exactly the overdue threshold (BR-FEE-008) — not only typical, mid-range records.
- Test data refresh/reset cadence between test cycles is Client/Product Decision Required.

12.1 Test Data Categories

Data Category	Purpose	Example
Baseline/Master Data	Populates reference entities so transactional testing has a valid foundation	Class, Section, Subject, FeeHead, Department (Appendix-G Master Data List)
Happy-Path Transactional Data	Verifies standard, expected-flow behavior	A student with complete profile, standard fee structure, normal attendance pattern
Boundary/Edge-Case Data	Verifies behavior exactly at a limit	A SeatAllocation at exactly RTE-quota ceiling; an Invoice exactly at the overdue threshold
Negative/Invalid Data	Verifies rejection behavior	A duplicate Aadhaar number; an out-of-range marks entry; a malformed mobile number
Scale/Volume Data	Supports Non-Functional Testing (Section 10)	20,000 Student records; 10 million+ AttendanceRecord entries, per Appendix-F Scalability targets

13. Defect Management Process

Defines how a defect is identified, logged, triaged, resolved, and closed — applied uniformly across all test levels (Section 8) and all 11 modules.

- Every defect is logged with: affected module (Section 8, Appendix-J Section 4), the specific Functional Requirement/Business Rule/NFR ID it violates (traceability per Section 17), Severity (Section 14), Priority (Section 15), steps to reproduce, and the environment in which it was found (Section 11).
- Defects affecting a Critical-priority Business Rule (Appendix-C v1.1, 19 of 71) or a Security NFR (Appendix-F) are escalated immediately, independent of the standard triage cadence, consistent with the Risk-Based Prioritization principle (Section 5).
- Root-cause categorization (requirement ambiguity, design gap, implementation error, environment/data issue) feeds the Entry Criteria review (Section 20) for the next test cycle, per the Defect Prevention principle (Section 5).
- A defect is never closed solely by the reporting QA engineer — resolution requires developer fix confirmation and independent QA retest (Section 16), consistent with the Independent Verification principle (Section 5).

13.1 Minimum Defect Report Fields

Field	Purpose
Defect ID	Unique identifier (Client/Product Decision Required for scheme)
Title/Summary	Concise description of the observed issue
Module & Component	Traces to Appendix-J Section 4 module and, where known, the specific internal component
Related FR/BR/NFR ID	Traces to Appendix-E, Appendix-C v1.1, or Appendix-F (Section 17 RTM linkage)
Severity / Priority	Per Sections 14-15
Environment	Per Appendix-L Section 4 (Development/Testing/Staging/Production)
Steps to Reproduce	Sufficient detail for independent reproduction
Expected vs. Actual Result	Clear statement of the discrepancy
Evidence	Screenshots, logs, or other supporting artifact

14. Defect Severity Matrix

Severity measures technical/functional impact, independent of business urgency (which is addressed separately by Priority, Section 15).

Severity	Definition	Representative Example
Critical	System unusable, data corruption, or a Critical-priority Business Rule (Appendix-C v1.1) is bypassed entirely	BR-FEE-001 (paid-invoice immutability) can be bypassed, allowing a locked invoice to be edited
High	Major functionality broken with no workaround, though the system remains otherwise usable	Marks entry (FR-18) fails for an entire subject, blocking the Examination module's downstream flow
Medium	Functionality impaired but a workaround exists	Export (FR-41) fails for one report format, but on-screen viewing and other formats remain functional
Low	Minor, cosmetic, or edge-case issue with negligible business impact	A validation message (Appendix-E Field 36) displays imprecise wording but the validation itself functions correctly

15. Defect Priority Matrix

Priority measures business urgency of resolution — how soon a defect, regardless of Severity, must be fixed.

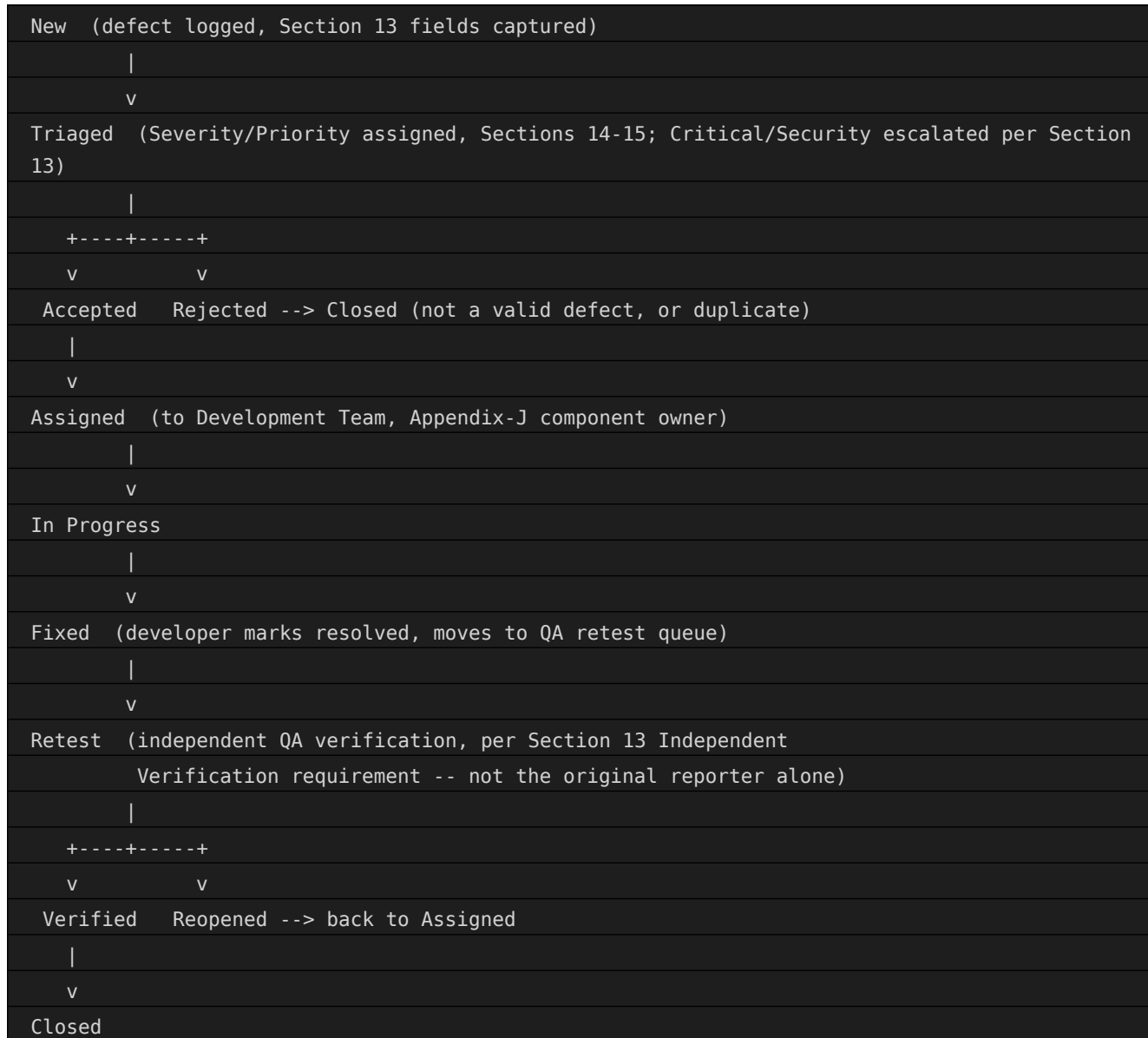
Priority	Definition	Typical Resolution Timeframe
P1 — Immediate	Blocks further testing or affects a live Production capability	Immediate — before any further test cycle proceeds
P2 — High	Must be fixed before the next Quality Gate (Section 25)	Within the current test cycle
P3 — Medium	Should be fixed before Production release, but does not block the current cycle	Within the release, before Production Quality Gate
P4 — Low	May be deferred to a future release with Project Sponsor agreement	Backlog, next release cycle

15.1 Severity-Priority Relationship

Severity and Priority are independent but correlated — a Critical-severity defect is typically also P1, but not always (e.g., a Critical-severity defect in a not-yet-released feature may be P2 if it does not block the current cycle's other testing). Both fields are recorded independently on every defect (Section 13).

16. Defect Life Cycle

Applied to every defect logged under Section 13, independent of Severity/Priority — every defect passes through the same state model.



17. Requirement Traceability Matrix (RTM)

This is a Test-Strategy-level RTM, distinct from and consistent with Appendix-B's Requirement Traceability Matrix — Appendix-B traces Functional Requirements to Business Rules, BPMN, and dependencies; this RTM traces those same Functional Requirements forward to test coverage, closing the loop from requirement to verified quality.

Functional Requirement (Appendix-E, e.g., FR-23)
v
Business Rule(s) (Appendix-C v1.1, e.g., BR-FEE-001)
v
Test Level (Section 8, e.g., System Testing + UAT)
v
Test Case ID (Client/Product Decision Required for the specific test-management tool/ID scheme)
v
Test Execution Result (Pass/Fail/Blocked)
v
Defect ID, if Fail (Section 13-16)

17.1 RTM Sample

Illustrative sample entries, reusing FR/BR IDs exactly as defined in Appendix-E and Appendix-C v1.1. A complete RTM covers all 47 Functional Requirements; this sample demonstrates the structure.

FR ID	Business Rule(s)	Test Level	Test Case ID	Status
FR-02	BR-ADM-002	System Testing, UAT	Client/Product Decision Required	Not Yet Executed
FR-10	BR-ATT-001, BR-ATT-002	System Testing, Regression	Client/Product Decision Required	Not Yet Executed
FR-18	BR-EXM-002, BR-EXM-003	System Testing, UAT	Client/Product Decision Required	Not Yet Executed
FR-23	BR-FEE-001	System Testing, UAT, Regression	Client/Product Decision	Not Yet Executed

FR ID	Business Rule(s)	Test Level	Test Case ID	Status
			Required	
FR-34	BR-HR-001, BR-HR-003	Integration, System Testing	Client/Product Decision Required	Not Yet Executed

Test Case IDs are marked Client/Product Decision Required since specific test-case numbering/tooling is an Implementation-phase decision; the FR-to-BR-to-Test-Level mapping itself is fully determined by Appendix-E and Appendix-C v1.1 and is not open.

18. Test Documentation

Document	Purpose	Produced At
Test Plan	Scope, schedule, and resourcing for a specific test cycle	Test Planning (Section 7)
Test Case Specification	Step-by-step verification of a specific FR/BR/NFR (Section 17 RTM)	Test Case Design (Section 7)
Test Execution Log	Record of each test case's execution result and evidence	Test Execution (Section 7)
Defect Report	Per Section 13 fields, for every logged defect	Throughout Test Execution
Test Summary Report	Consolidated results, metrics (Section 24), and Quality Gate recommendation (Section 25)	Test Closure (Section 7)
RTM	Per Section 17, updated continuously through the cycle	Continuous, finalized at Test Closure

18.1 Documentation Ownership

Test Plans and Test Case Specifications are owned and maintained by the QA Team, consistent with the QA Team's primary responsibility for Integration, System, and Regression Testing (Section 6 table). Test Summary Reports and the RTM (Section 17) are jointly owned by the QA Team and reviewed by the Project Sponsor before each Quality Gate (Section 25), ensuring the same Separation-of-Duties discipline already established in Appendix-L Section 22.2 applies to test governance, not only to Production access.

19. Test Deliverables

Deliverable	Description	Recipient
Test Strategy (this document)	Overall approach, levels, and quality gates	Project Sponsor, Development Team, QA Team
Test Plan per cycle	Cycle-specific scope and schedule	QA Team, Development Team
Test Case Repository	Full set of test cases traced per Section 17	QA Team
Defect Log	All logged defects with current Section 16 lifecycle state	Development Team, QA Team, Project Sponsor
Test Summary Report per cycle	Results, metrics, Quality Gate recommendation	Project Sponsor (Appendix-I Section 8), Management
Final UAT Sign-off	Client confirmation per Section 22 Acceptance Criteria	Project Sponsor, per RGD v2.0 Approval Matrix

19.1 Deliverable Timing

Deliverable	Timing Relative to Test Cycle
Test Strategy (this document)	Once, before the first test cycle begins
Test Plan per cycle	Before Entry Criteria (Section 20) are evaluated for that cycle
Test Case Repository updates	Continuous, as new FR/BR/NFR coverage is designed
Defect Log	Continuous throughout Test Execution (Section 7)
Test Summary Report	At Test Closure (Section 7), before the corresponding Quality Gate (Section 25)
Final UAT Sign-off	At Gate 3 (Section 25), before Production promotion

20. Entry Criteria

Conditions that must be satisfied before a given test level (Section 8) may begin — prevents wasted test effort against an unready build or environment.

Test Level	Entry Criteria
Unit Testing	Component code complete per Appendix-J Section 4 design; Development environment available (Appendix-L Section 4)
Integration Testing	Unit Testing passed for all involved components; Controller-Service-Repository interfaces stable (Appendix-J Section 6)
System Testing	Integration Testing passed; Testing/QA environment ready (Appendix-L Section 4); test data available (Section 12)
User Acceptance Testing	System Testing exit criteria met (Section 21); Staging environment ready and production-representative (Appendix-L Section 4.2)
Regression Testing	A code/configuration change has been made; affected-module scope identified via Appendix-D dependency chain
Smoke Testing	A new deployment has completed (Appendix-L Section 19) in the target environment
Production Verification	Production deployment complete; Smoke Testing passed in Staging beforehand

21. Exit Criteria

Conditions that must be satisfied before a test level is considered complete and the next may begin, or before a Quality Gate (Section 25) may be evaluated.

Test Level	Exit Criteria
Unit Testing	100% of designed unit test cases executed; no Critical/High severity defects open
Integration Testing	All Appendix-J Section 6/13 interaction points and cross-module events verified; no Critical severity defects open
System Testing	100% of Section 17 RTM entries executed at least once; all Critical-priority Business Rules (Appendix-C v1.1) verified; no Critical/High severity defects open
User Acceptance Testing	All RGD v2.0/Appendix-E business scenarios executed; Client sign-off obtained (Section 22)
Regression Testing	All previously-passing test cases in the affected scope re-executed and passed
Non-Functional Testing (Section 10)	All applicable Appendix-F NFR targets met or formally waived by Project Sponsor with documented rationale

21.1 Release Validation Flow

Conceptual flow specifically for validating a release candidate before it is promoted from UAT/Staging to Production, reusing the Deployment Lifecycle already established in Appendix-L Section 19.



v
Acceptance Criteria checklist completed (Section 22)
v
Client sign-off obtained (RGD v2.0 Approval Matrix)
v
Release candidate approved for Production (Gate 3, Section 25)

22. Acceptance Criteria

Reuses the Gherkin-style Given/When/Then acceptance criteria already defined per Functional Requirement in Appendix-E Field 26 — this section defines the aggregation rule for overall system acceptance, not new per-requirement criteria.

Acceptance Checklist Item	Basis
100% of 47 Functional Requirements' Appendix-E Field 26 acceptance criteria pass	Appendix-E
100% of 19 Critical-priority Business Rules verified with zero open Critical-severity defects	Appendix-C v1.1 Section 2.2
All applicable Appendix-F NFR targets met or formally waived	Appendix-F
Zero open Critical or High severity defects (Section 14) at time of sign-off	Section 14
Client (Project Sponsor) UAT sign-off obtained, per RGD v2.0 Approval Matrix	RGD v2.0
Disaster Recovery drill (Appendix-L Section 13.1) successfully rehearsed in Staging at least once	Appendix-L Section 19.4/13

22.1 Acceptance Sign-Off Process

Client acceptance is not a single, all-or-nothing event — it is evaluated against the checklist above item by item, with any unmet item either resolved before sign-off or explicitly waived in writing by the Project Sponsor with documented rationale, consistent with the Conditional Pass outcome already defined for Quality Gates (Section 25). A waived item is tracked forward into Production Verification (Section 8) monitoring, not simply forgotten once sign-off is granted.

23. Risk Based Testing

Directly operationalizes the Risk-Based Prioritization principle (Section 5), reusing risk data already established in Appendix-D, Appendix-C v1.1, and Appendix-I/J/K/L.

Risk Area	Source	Testing Response
Fee Collection module (highest financial risk)	Appendix-I Section 5.1.6	Highest test-case density; independent verification mandatory (Section 5) for all Fee-related Critical Business Rules
BR-ADM-001/BR-ADM-007 seat-allocation race condition	Appendix-C v1.1 Section 3.2; Appendix-H v1.1 Section 19	Dedicated concurrency/race-condition test scenarios, not covered by standard functional test cases alone
FR-37 Communication as cross-module single point of failure	Appendix-D Dependency Risk Register; Appendix-I Section 26.2	Integration Testing specifically includes per-channel failure-isolation scenarios (Appendix-I Section 11.2)
Polymorphic foreign-key integrity	Appendix-H v1.1 Section 19; Appendix-J Section 22	Dedicated data-integrity test scenarios for User, Document, ApprovalRequest, NotificationLog, BookIssue
Statutory/regulatory compliance (RTE, DPDP, GST, payroll)	Appendix-C v1.1 regulatory-mapped rules; RGD v2.0 Section 16	Compliance-focused test scenarios independently reviewed, not solely developer-verified

23.1 Test Effort Allocation

Consistent with the Fee Collection and Admission modules' elevated Business Rule density already noted in Appendix-I Section 5.1, test-case allocation across the 11 modules is weighted toward these two modules and toward HR & Payroll (statutory compliance risk), rather than allocated evenly per module regardless of risk profile.

23.2 Test Intensity by Module Risk Tier

Risk Tier	Modules	Test Intensity
Highest	Fees, Admission, HR & Payroll	Maximum test-case density; independent verification mandatory for all Critical Business Rules; dedicated Non-Functional Testing pass
Elevated	Student Information, Examination, Attendance	High test-case density given their central/gating role in the module dependency chain (Appendix-D)
Standard	Timetable-adjacent flows, Transport, Communication, Administration	Standard test-case density per Appendix-E acceptance criteria
Lower	Library, Reports	Proportionate test-case density given lower Business Rule count (Appendix-C v1.1 Section

Risk Tier	Modules	Test Intensity
		2.1) and read-mostly nature (Reports, BR-RPT-002)

24. Test Metrics

Objective measures reported at each Test Summary Report (Section 18) and used as Quality Gate (Section 25) input.

Metric	Definition	Target
Requirement Coverage	% of 47 Functional Requirements with ≥ 1 executed test case (Section 17 RTM)	100% before UAT exit
Business Rule Coverage	% of 71 Business Rules with ≥ 1 executed test case	100% before System Testing exit, Critical rules first
Test Case Pass Rate	% of executed test cases passing	Client/Product Decision Required (target threshold per cycle)
Defect Density	Defects found per module or per Functional Requirement	Tracked for trend, no fixed target — informs Section 23 risk weighting for future cycles
Defect Escape Rate	Defects found in a later test level that should have been caught earlier (Section 5 Shift-Left)	Trending toward zero over successive cycles
Critical/High Defect Aging	Time from defect logged (Section 13) to Verified/Closed (Section 16)	Client/Product Decision Required (target resolution SLA)
NFR Compliance Rate	% of applicable Appendix-F NFR targets met	100% before Production Quality Gate, or formally waived

24.1 Metric Reporting Cadence

Requirement Coverage, Business Rule Coverage, and Test Case Pass Rate are reported at the end of every test cycle (Section 7 Test Closure), feeding directly into the corresponding Quality Gate (Section 25). Defect Density and Defect Escape Rate are trended across cycles, not evaluated in isolation per cycle, since their value lies in identifying whether the Shift-Left principle (Section 5) is actually taking effect over time.

25. Quality Gates

Formal checkpoints gating progression through the environments (Appendix-L Section 4.1) — a Quality Gate decision is Pass, Conditional Pass (with documented exceptions), or Fail.

Gate 1: Development -> Testing/QA		
Entry: Unit Testing exit criteria met (Section 21)		
v		
Gate 2: Testing/QA -> UAT/Staging		
Entry: System Testing exit criteria met; RTM Requirement		
Coverage = 100% (Section 24)		
v		
Gate 3: UAT/Staging -> Production		
Entry: Acceptance Criteria met (Section 22); Client sign-off		
obtained; Non-Functional Testing exit criteria met		
v		
Gate 4: Production Release Confirmation		
Entry: Post-release Smoke Testing passed (Section 8);		
Monitoring (Appendix-L Section 17) confirms stability		
Gate	Decision Authority	Possible Outcomes
Gate 1	QA Lead	Pass / Fail (Fail returns to Development)
Gate 2	QA Lead + Development Lead	Pass / Conditional Pass (documented open Medium/Low defects) / Fail
Gate 3	Project Sponsor (per RGD v2.0 Approval Matrix)	Pass / Conditional Pass (documented NFR waivers) / Fail
Gate 4	Project Sponsor + Infrastructure/DevOps (Appendix-L Section 22)	Pass / Rollback (per Appendix-L Section 21 Rollback Strategy)

26. Final Validation Checklist

Check	Result
All 27 required sections present	PASS — Sections 1 through 27 as specified
All 7 Test Levels detailed (Section 8)	PASS — Unit, Integration, System, UAT, Regression, Smoke, Sanity
All 8 Non-Functional Testing categories detailed (Section 10)	PASS — Performance, Scalability, Reliability, Availability, Security, Usability, Accessibility, Compatibility
All 4 environments mapped (Section 11)	PASS — Development, Testing, UAT/Staging, Production Verification, consistent with Appendix-L Section 4
No source code, SQL, APIs, JSON/XML, or test scripts included	PASS
No tool-specific artifacts (Selenium, Postman, JMeter, Cypress, Playwright) included	PASS
Required ASCII diagrams present (6 of 6)	PASS — Testing Life Cycle, Defect Life Cycle, Requirement Traceability Flow, Test Environment Flow, Release Validation Flow, Quality Gate Flow
Required tables present (11 of 11)	PASS — Test Level Comparison, Test Types, Severity Matrix, Priority Matrix, RTM Sample, Entry Criteria, Exit Criteria, Acceptance Checklist, Test Deliverables, Test Metrics, Quality Gates
No new modules, database objects, APIs, business rules, or infrastructure invented	PASS — every reference traces to Appendix-F/H/I/J/K/L by ID
Every unknown testing decision flagged	PASS — marked Client/Product Decision Required consistently with prior appendices

27. Document Closure

Metric	Value
Test Levels Defined	7 of 7 requested
Non-Functional Testing Categories Covered	8 of 8 requested
Required ASCII Diagrams	6 of 6 (Testing Life Cycle, Defect Life Cycle, Requirement Traceability Flow, Test Environment Flow, Release Validation Flow, Quality Gate Flow)
Required Tables	11 of 11 (Section 8, 9, 14, 15, 17.1, 20, 21, 22, 19, 24, 25)
Quality Gates Defined	4 (Development→Testing, Testing→UAT, UAT→Production, Production Release Confirmation)
Traceability to Appendix-F/H/I/J/K/L	Complete — reused by ID throughout
No Fabricated Scope	Confirmed — every module, entity, rule, API, and infrastructure element reused by reference
Test Strategy & QA Quality Score	91/100
Approval Status	Conditionally Ready — Test Strategy is structurally complete across all 27 sections, 7 test levels, and 8 NFR categories; approval is gated on the consolidated Client/Product Decision Required items already tracked across Appendix-C v1.1, F, G, H v1.1, I, J, K, and L, plus test-specific open decisions in Sections 6, 12, 17.1, and 24 (automation approach, test data refresh cadence, test-case ID scheme, and metric target thresholds)

Deductions from a perfect score reflect the volume of tooling and threshold decisions that remain legitimately open pending client/product input, not any structural incompleteness in this document's own 27 sections, 6 diagrams, or 11 tables. This document formally closes Gap-13 and awaits review before any subsequent appendix is considered.

27.1 Consolidated Documentation Chain

This Test Strategy is the thirteenth document in an unbroken chain, each building on the last without contradiction: RGD v2.0 established business requirements; Appendix-A through D established process, traceability, dependency, and business-rule detail; Appendix-E and F established functional and non-functional requirements; Appendix-G and H v1.1 established the data model; Appendix-I established system architecture; Appendix-J established internal component design; Appendix-K established the external interface shape; Appendix-L established deployment infrastructure; and this Appendix-M establishes how all of the above will be verified for quality. No single document in this chain can be understood in isolation from those preceding it, and this document has deliberately avoided restating their content rather than merely referencing it.